NORTHLIGHT RESEARCH

The State of the Page

How a 30-year-old format keeps getting better

ANNUAL REPORT · 2026 EDITION

Not for redistribution — a sample document

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Every entry above is a link annotation with a GoTo destination, and the same four sections are listed in the document outline (bookmarks panel).

Executive summary

KEY MESSAGE

Document technology rarely makes headlines, yet it quietly carries contracts, invoices, manuals and archives for billions of people. This report reviews how a mature page-description format stays reliable across decades of devices, operating systems and software generations.

Three forces explain its longevity: a small, fully specified imaging model; a conservative file structure that tolerates partial damage; and an ecosystem of independent implementations that keep each other honest. Each is examined in the sections that follow.

The findings are drawn from a corpus of 4,812 public documents and from the behaviour of four independent renderers. Where results differ we say so, and where a difference is a bug we link it to the clause of the specification that decides the matter.

> *“Longevity is a design choice, not an accident of adoption.”*
>
> — Dr. Imani Okafor, lead author

4,812	97.3 %	12	30 yrs
documents analysed	rendered identically	format clauses clarified	of backward compatibility

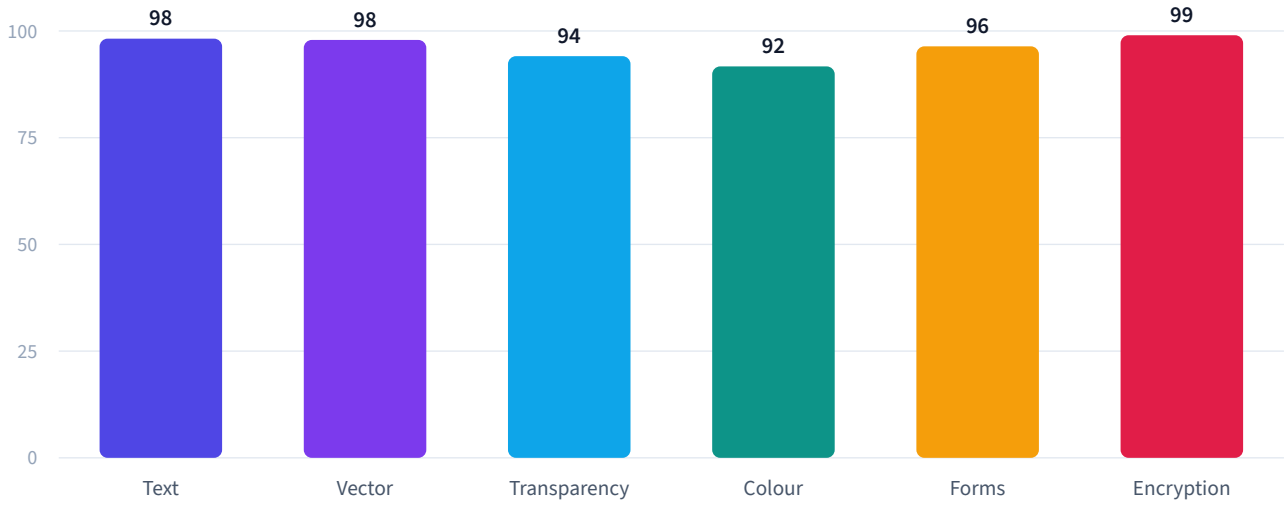
Findings

Compatibility improved across every feature family we measured. The table lists the share of corpus documents that used each feature and the share rendered without visible difference.

FEATURE FAMILY	USAGE	IDENTICAL
Text and fonts	99.1 %	98.2 %
Vector graphics	87.4 %	97.9 %
Transparency	41.6 %	94.1 %
Colour management	33.8 %	91.7 %
Forms and annotations	22.5 %	96.4 %
Encryption	9.3 %	99.0 %

FIGURE 1

Identical rendering by feature family, %

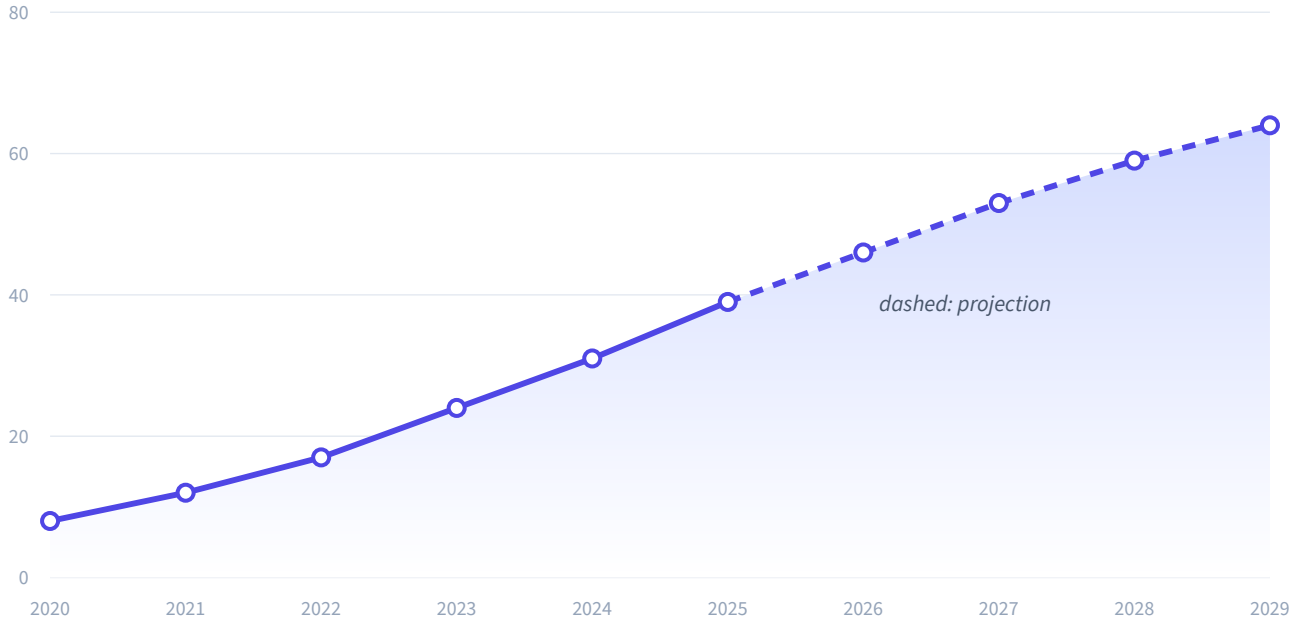


Outlook

Adoption of tagged, accessible documents is the clearest growth story. If the current trend holds, a majority of new documents will carry a structure tree by 2029.

FIGURE 2

Share of new documents with a structure tree, %



Accessible by default

Authoring tools tag structure automatically.

Smaller files

Object streams and subsetting cut size by a third.

Safer by design

AES-256 is now the baseline for protected files.

Method and sources

Documents were sampled from public repositories, converted to a common baseline where necessary, and rendered at 96 dpi by four independent engines. A document counts as identical when no channel differs by more than 2 % over more than 2 % of pixels.

REFERENCES

ISO 32000-2:2020	Document management — Portable document format — Part 2: PDF 2.0
ISO 14289-1:2014	PDF/UA-1 — accessible PDF
ISO 19005-4:2020	PDF/A-4 — archival
RFC 8118	The application/pdf media type

Colophon

Set in Source Serif 4 and Source Sans 3 (embedded CFF subsets). Charts are drawn with path operators and shadings; there are no raster images in this file.